



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Wartość informacyjna danych rzeczywistych w uczeniu robotów w pętli rzeczywistość–symulacja–rzeczywistość: metoda ograniczania ich ilości z zastosowaniem w nawigacji i manipulacji

EN: The value of real data in real-to-sim-to-real robot learning: a method for reducing its amount, with applications to navigation and manipulation

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),



- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	(Mar – Sep 2026) Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the data-efficient real-to-sim-to-real thesis, research questions RQ1–RQ4 (capture less, learn robustly in an imperfect twin, few real data, budget of the whole loop) and hypotheses H1–H4 with measurable decision rules and a proxy-reality evaluation protocol. <i>D</i>: §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D</i>: tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D</i>: this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	(Oct 2026 – Feb 2027) Mechanism 1 – less capture (RQ1, H1). T3.1 Proxy-reality set-up (§9): application for the ScanNet++ licence (signed by the supervisor) and for a PLGrid computing grant in Oct 2026; ≥ 20 ScanNet++ scenes (10 held out) and the MuSHRoom rooms, with laser-scan or mesh references and separate phone captures; twins on the budget grid with the open simulator, 3DGS and structure-from-motion tools; hidden physical parameters of "reality"; navigation tasks, learner and pretrained models running on the twins; pilot fixing the difficulty regime, the operator-minute cost model, the sim-vs-real correlation between twin and "reality" and the GPU cost; pre-registration of the mechanism 1 baselines. <i>D</i>: working pipeline and pre-registered protocol in the repository, by Jan 2027. Place: K46, WCSS. T3.2 Validation stand: SzD Minigrant application for a small mobile robot, or a written agreement on robot access with the K29 Denali laboratory. <i>D</i>: application or agreement, by Nov 2026. T3.3 Mechanism 1: capture guided by the task and by the twin's uncertainty vs. uniform, task-blind and task-weighted selection, judged by the success of the model trained in each twin. <i>D</i>: first budget–success curves (first H1 result), by Feb 2027.



Semestr 4	(Mar – Sep 2027) Mechanism 1 completed; mechanism 2 – better use of simulation (RQ1–RQ2, H1–H2). T4.1 Mechanism 1 completed: full budget curves against all pre-registered baselines. <i>D:</i> H1 completed on the proxy reality and an early preprint of the result, by Apr 2027. T4.2 Prepare and submit article P1 to NeurIPS 2027 (mechanism 1, H1, in navigation). <i>D:</i> submission, May 2027 (§11). T4.3 Mechanism 2: training that follows the twin's uncertainty, initialized from pretrained foundation models, vs. uniform and no domain randomization in the pre-registered difficulty regime; the world-model extension as an ablation. <i>D:</i> first H2 result by Jun 2027; H2 completed with the ablations, by Jul 2027. Place: K46, WCSS. T4.4 Mechanism 3 set-up: gap prediction, selection and correction code on the proxy reality, check of the sim-vs-real precondition (Jul–Sep 2027). <i>D:</i> first H3 result, by Sep 2027. T4.5 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D:</i> applications, poster.
Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Mechanism 3 – fewer real trials (RQ3, H3); manipulation set-up. T5.1 Mechanism 3: selection of real trials by the predicted twin-to-reality gap vs. random, uniform-coverage and failure-driven selection; correction of the twin (appearance, geometry, physical parameters) and of the model with the selected trials. <i>D:</i> H3 completed on the proxy reality, by Jan 2028. Place: K46, WCSS. T5.2 First robot validation campaign (2 PWr rooms, ~8 robot-hours): own captures, twins, uniform capture vs. the method, sim-vs-real correlation between proxy and robot. <i>D:</i> validation report. Place: K46 or Denali. Without a robot, the campaign moves to sem. 7; H1–H3 are decided on the proxy reality either way. T5.3 Generalization set-up: manipulation proxy reality with hidden physical parameters (§9) and mechanisms 1–2 run on it unchanged. <i>D:</i> first manipulation budget curves, by Dec 2027. T5.4 Article P2 (mechanisms 2–3, RQ2–RQ3, in navigation, with the first manipulation results) submitted to ICML 2028 (~late Jan 2028; CVPR 2028, ~Nov 2027, only if H3 is complete by then). If P1 was rejected, its revision goes to CVPR 2028. <i>D:</i> submission(s).



Semestr 6	(Mar – Sep 2028) The whole method (RQ4, H4, the thesis) and its generalization. T6.1 Budget curves of the whole method vs. the assembled, pre-registered strongest existing real-to-sim-to-real approach and vs. learning from real data only, with the method's settings fixed in advance; the whole method, unchanged, in manipulation. <i>D</i>: first H4 result by Apr 2028; manipulation curves by Jul 2028; H4 completed, by Sep 2028. Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D</i>: visit report. T6.3 Article P3 (the whole method against existing approaches, with the generalization to manipulation) submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). <i>D</i>: submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign of the whole method; final analysis of H1–H4. <i>D</i>: final results. Place: K46, Denali. T7.2 Release of the method's code, configurations, scene lists, seeds, pre-registrations and the releasable twins (no assets derived from ScanNet++). <i>D</i>: public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D</i>: submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D</i>: dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D</i>: revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).



Robots that learn a task from experience, such as moving around a building or handling objects, need a lot of that experience from the place where they will work, and collecting it in reality is slow, expensive and sometimes unsafe. Training in simulation avoids this cost but opens a **sim-to-real gap**: a model that works in a generic simulator often fails in the real place. **Digital twins** built from real data narrow this gap. Neural scene reconstruction turns a short video of a real scene into a photorealistic model of its geometry and appearance, and a few measurements give the physical properties that matter for the task. This gives the **real-to-sim-to-real loop**: acquire real data, build a twin, train the robot's model in it at scale, and transfer it back to reality, where a little more real data checks and corrects the twin and the model. Each part of the loop is an active research area, and twins built from a phone video already train navigation and manipulation models that work in reality.

What remains open is **how much real data the loop needs, and how to need less of it**. In every existing pipeline the amount is fixed by hand: the scene is captured uniformly, training uses generic randomization, and real trials, if any at all, are chosen at random or by hand. To our knowledge, no navigation pipeline has a stage that corrects the twin with real trials, and none reports how success depends on the real data spent; in manipulation such accounting exists only at single points.

Scientific problem and goal. The underlying problem is scientific: learning under distribution shift when target-domain data can be bought at a cost at several points of a learning loop. The dissertation poses the real-to-sim-to-real loop as sequential experimental design over a real-data budget, values each real datum by its expected reduction of the twin-to-reality performance gap per unit cost, and derives when twin data can replace real data. The goal is **a method, grounded in this principle, that reduces the amount of real data** needed to teach a robot a task, with one mechanism per step: (1) **less capture**; (2) **better use of simulation**, training so that the twin's errors do not transfer; (3) **fewer real trials**, choosing those that correct twin and model most. The thesis is that the method needs at least two times less real data than the strongest existing real-to-sim-to-real approach. Operator minutes are only the evaluation measure. Indoor navigation is the main testbed; manipulation tests whether the same method carries over.

Why this topic, and why in this discipline. The object is a data-efficient learning method (active learning, learning under distribution shift, uncertainty estimation, adaptation of pretrained models), not a particular robot, simulator or benchmark, which places it in *information and communication technology*. The student builds on existing open reconstruction pipelines, simulators, pretrained models and public datasets. Public scans of real buildings make it possible to count real data exactly without an own robot; trials on a small mobile robot validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning, and the training runs on the GPU infrastructure available to PWr researchers (WCSS, PLGrid).

Potential application areas: faster and cheaper deployment of robots in new places, such as warehouses, hospitals, offices and inspection sites, with less time spent collecting data and supervising real trials; guidance on how much real data to collect for a twin; and, beyond robotics, any model trained on reconstructed or simulated data and used on real sensor data.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).



Simulation and sim-to-real transfer. Robot models for navigation and manipulation are commonly trained in simulators such as Habitat [1] and ManiSkill3 [2], where large-scale reinforcement learning essentially solves point-goal navigation after billions of steps [3]. Transfer to the real world is the open part: simulation success need not predict real success (the Sim-vs-Real Correlation Coefficient rose from 0.18 to 0.84 once one collision artefact was removed [4]; paired evaluations in manipulation [5]). Domain randomization varies appearance [6] or dynamics [7] uniformly within hand-set ranges, and a few real rollouts can adapt the simulation parameter distribution [8]. In navigation, more data from a known location saturates quickly [9]; sim-and-real co-training raises real success [10].

From real data to digital twins. Neural scene reconstruction, above all 3D Gaussian Splatting (3DGS) [11], turns a short real capture into a photorealistic model of a scene. Twins built this way train models that transfer. In navigation, EmbodiedSplat [12] reconstructs rooms from a phone capture and fine-tunes image-goal navigation in Habitat, Vid2Sim [13] builds simulators for urban navigation from monocular video, GaussGym [14] trains from pixels, and ReaDy-Go [15] adds moving obstacles. In manipulation, RialTo [16] builds a twin from a scan and a few demonstrations, X-Sim [17] from a minute of human video, and TwinRL [18] fine-tunes a pretrained policy in a phone-captured twin and targets its few real rollouts at configurations where the twin predicts failure. The physical part of a twin can be identified from a small amount of real data, as a posterior over simulator parameters [19] or by exploration designed for identification [20]. Datasets such as ScanNet++ [21] pair laser scans and DSLR images with a separate phone capture of the same scene, and MuSHRoom [22] pairs phone and depth-camera captures with a reference mesh under an open licence. In all these works the amount of real data is fixed by hand: the navigation twins use between 15 seconds and 30 minutes of video per scene and no real training trials, and the manipulation works report single points (e.g. a scan and 15 demonstrations instead of 50 demonstrations [16]) rather than curves.

Capturing the right real data. Active view selection chooses the images that improve a reconstruction most: FisherRF [23] maximizes expected information gain, and Bayes' Rays [24] estimates an uncertainty field for a trained radiance field. Risk-aware view acquisition [25] weights FisherRF by safety-critical regions for navigation, and AREA3D [26] adds vision-language guidance. None measures the success of a model trained in the resulting twin (at most two task-aware capture papers a year, 2023–2026).

Foundation models, world models and real trials. Pretrained navigation foundation models generalize across robots and environments (GNM [27], ViNT [28]), and vision-language-action models map images and instructions to actions [29]; they still need data from the target place to work there. Learned world models predict future observations: navigation world models plan by imagining trajectories [30], and open world foundation models are post-trained per setup [31]; VLAW [32] improves a world model with real rollouts, at a fixed, hand-chosen number of them. Choosing real trials has been studied for *evaluating* policies (a fifth to a quarter less robot effort [33]), not for *correcting* a twin and its model. Frozen visual encoders [34] make twin and real observations comparable, and prediction-powered inference

[35] estimates real performance from many cheap predictions and a few real labels.

Theory. Domain-adaptation theory bounds target error by source error plus a discrepancy between domains [38], and Bayesian experimental design chooses measurements by expected information gain [36, 37]; neither has been applied to the allocation of real data across a real-to-sim-to-real loop.

Research gap. Real-to-sim-to-real learning works, but the real data it consumes is set by hand and spent uniformly at every step. No navigation pipeline has a stage that corrects the twin with real trials, and none reports how success depends on the real data spent; in manipulation the real data is accounted at single points, against learning from real data only, never against another real-to-sim-to-real pipeline. Missing are a principled criterion for spending real data across the whole loop, its theory, and a method built on it that reduces real data at every step (RQ1–RQ3) and in total (RQ4).

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Scientific problem. Learning with a digital twin is learning under distribution shift in which the learner can *buy* target-domain (real) data at a cost, at three points of the real-to-sim-to-real loop: to build the twin, to train in it and to correct the model in reality. How to allocate a scarce real-data budget across these points so that the loss of real-world performance is smallest is an open problem of sequential experimental design [36, 37] that has not been posed for the whole loop.

Principle and theoretical claim. The loop is formalised as sequential Bayesian experimental design in which each real datum (a view, an interaction, a trial) is valued by its **expected reduction of the twin-to-reality performance gap per unit cost**. Following domain-adaptation theory [38], the real performance of a model trained in the twin is bounded by its twin performance plus a discrepancy term on the states the task visits; the dissertation derives a task-weighted, uncertainty-based



estimate of this term and the conditions under which spending real data on it pays off. The three mechanisms of the method, (1) less capture, (2) better use of simulation, (3) fewer real trials, are instances of this one criterion.

Thesis. Allocating real data by its expected gap reduction, rather than uniformly or by hand, makes the loop at least twice as economical in real data as the strongest existing real-to-sim-to-real approach.

How the hypotheses are decided. Real effort is counted in **operator minutes** on one budget grid; P is the success rate in "reality" and the target P a fraction of the baseline's plateau. Decisions are made on a non-circular proxy reality from public scans of real buildings, with pre-registered baselines; a real robot validates the direction and the proxy. Navigation decides H1–H4; manipulation tests the same method, unchanged. Details are in §9.

RQ1 (step 1). Which real data is worth acquiring to build the twin? **H1.** Capture chosen by the criterion reaches the P of uniform capture with **at least 40% less capture**.

RQ2 (step 2). How should the model be trained so that the twin's errors do not transfer? **H2.** At the same budget, training weighted by the twin's uncertainty gives a P **at least 10 pp higher** than uniform domain randomization, in a pre-registered regime where the baseline succeeds in $\leq \sim 75\%$ of episodes; a learned world model filling the twin's gaps is an ablation.

RQ3 (step 3). Which few real trials correct the twin and the model best? **H3.** Trials chosen by the predicted gap reach the target P with **at least 50% fewer trials** than random ones, and fewer than a failure-driven rule, if the twin predicts reality well enough (minimum sim-vs-real correlation, §9).

RQ4 (theory and the whole loop). When and why can twin data replace real data? **H4 (thesis).** The whole method reaches the target P with **at least 2× less real data** than the strongest existing approach (assembled, pre-registered, same budget), and the savings follow the conditions predicted by the theory: they grow with the twin-to-reality correlation and with how concentrated the gap is. In manipulation the effects keep their sign.

A hypothesis that fails is reported as a measured curve, which still answers its question.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is in **information and communication technology** (*informatyka techniczna i telekomunikacja*), area machine learning. Its object is the **real-to-sim-to-real loop of robot learning and the real data it consumes**: transferring real-world data into a digital twin, training a model in it and transferring the model back to reality. Building simulators, world models, foundation models or benchmarks from scratch, control design and mechanical engineering are not the object of the research; the student uses existing tools, pretrained models and public data.

Key original contribution: a scientific answer to how real data should be spent. (i) A formulation of the real-to-sim-to-real loop as sequential experimental design over a real-data budget, with one criterion (expected reduction of the twin-to-reality performance gap per unit cost) and a domain-adaptation-style bound [38] that states when twin data can replace real data; (ii) a method that instantiates this criterion at every step of the loop and reaches a given real-world performance with at least two times less real data than the strongest existing approach (RQ4, H4); (iii) empirical knowledge of when and why the savings arise. What is new is not the twin but the principled



spending of real data. The three mechanisms are instances of the criterion, each tested at its own step:

1. **Less capture (RQ1, H1).** Capture of real data guided by what matters for the task and by where the twin is still uncertain, for appearance, geometry and the physical properties the task needs; compared with uniform capture and with task-blind and task-weighted selection aimed at reconstruction quality.
2. **Better use of simulation (RQ2, H2).** Training that follows the twin's uncertainty, so that its errors do not transfer, with a learned world model filling the twin's gaps; compared with uniform and with no domain randomization.
3. **Fewer real trials (RQ3, H3).** Selection of the few real trials that the predicted twin-to-reality gap marks as most informative, used to correct both the twin and the model; compared with random and with failure-driven selection.

Generalization. H1–H4 are decided in indoor robot navigation; the same method, unchanged, is then run in robotic manipulation as a generalization test, with no separate thresholds.

How it is evaluated and released (supporting, not contributions in their own right). Curves of success against operator minutes on a shared budget grid, against pre-registered baselines and learning from real data only; a proxy-reality protocol on public scans of real buildings that counts real data exactly without a robot (a separate, higher-fidelity reference with hidden physical parameters plays "reality"), validated against a real robot; and open code, configurations, scene lists, seeds and pre-registrations, built on existing tools.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems; a method that decides where it is worth spending, and a measured estimate of how much of it a twin can replace, are relevant to active learning, learning under distribution shift, uncertainty estimation and adaptation of pretrained models in general, beyond robotics.

Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS, ICML and ICLR (machine learning) and CVPR (computer vision), with ICCV and ECCV for resubmissions. P1 (mechanism 1, RQ1, in navigation) targets NeurIPS 2027; P2 (mechanisms 2–3, RQ2–RQ3, in navigation, with first manipulation results) ICML 2028 or CVPR 2028; P3 (the whole method against existing approaches, RQ4, with the generalization to manipulation) NeurIPS 2028, with ICLR 2029 as the fallback.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The dissertation develops **one method** with three reduction mechanisms, one per step of the real-to-sim-to-real loop: less capture, better use of simulation, fewer real trials. Each mechanism is developed and tested at its own step (RQ1–RQ3); the whole method is then tested against the strongest existing approach (RQ4). Existing open tools that run on the data-centre GPUs of WCSS and PLGrid are used throughout (Habitat-Sim/Lab [1], an open 3DGS library, COLMAP); nothing is built from scratch.

Theoretical part (RQ4; sem. 3–5, then refined). The loop is written as sequential Bayesian experimental design [36, 37]: a posterior over the twin (geometry, appearance, physical parameters) and over the model's twin-to-reality gap, updated by each real datum. A task-weighted discrepancy bound in the spirit of [38] links the twin's uncertainty on the states the task visits to the loss of real performance; the acquisition criterion (expected gap reduction per operator minute) follows from it and is used at all three steps. The predictions of the analysis, when each mechanism saves data and how much, are tested against the measured budget curves (H4).



Real data: one unit, one grid. All real effort is counted in **operator minutes** B = capture + demonstrations + on-robot trials with resets; raw counts (frames, trials) are reported. Every approach gets the same budget grid, e.g. $B \in \{5, 10, 20, 40, 80\}$ min, the same initialization, compute in the twin and test episodes. Each comparison yields the budget at which an approach reaches the target P , with a scene-level bootstrap interval for the budget ratio. The target P is a fraction of the baseline's plateau (e.g. 90% of its P at the largest budget), so "40%" and "2×" stay well defined when the baseline's curve flattens.

Proxy reality (decides H1–H4). On ≥ 20 scenes of ScanNet++ [21] (10 held out; licence signed by the supervisor; derived twins not published) and the rooms of MuSHRoom [22] (open licence; twins releasable), "reality" is the laser-scan mesh textured from the DSLR images and rendered in Habitat with a robot-camera model, with its own physical parameters (e.g. actuation noise, friction) hidden from the method and learned only from counted real data; this physical gap is synthetic, and the robot checks it. The twin is built from a subset of the separate phone capture of the same scene, so twin and "reality" share neither images nor reconstruction method; the largest budget stays below the full capture. Tasks are goal-reaching navigation (image-goal and point-goal) in static indoor scenes, with the difficulty set in a pilot. The sim-vs-real correlation [4] between twin and "reality" is measured per scene and must exceed a pre-registered threshold for H3. The proxy's own error is reported, and every hypothesis is re-checked for sign with a second reference from all DSLR images. P = success rate; path efficiency is also reported.

Mechanism 1 – less capture (RQ1, H1; sem. 3–4).

- *Method:* the twin's appearance and geometry come from 3DGS [11], run as a mesh in the simulator (as in [12]); its physical part is identified from a few trajectories as a posterior [19]. The next views are chosen where the twin's uncertainty (Fisher information [23] or an uncertainty field [24] adapted to 3DGS) is highest in the regions that matter for the task, e.g. along planner paths between likely goals; one criterion covers views and physical parameters.
- *Pre-registered baselines:* uniform capture; task-blind information-driven selection [23]; risk- and semantics-weighted selection with a reconstruction objective [25, 26]; task-blind identification of the physical part [20].
- *Criterion (H1):* $\geq 40\%$ less capture than uniform at the target P (same learner in each twin; reference budget below the plateau), upper 95% bound of the ratio below 1; against task-blind selection only the upper bound below 1 is claimed.

Mechanism 2 – better use of simulation (RQ2, H2; sem. 4).

- *Method:* the model is initialized from a pretrained foundation model (e.g. a general navigation model or a vision-language-action model [27–29]) and trained in the twin by imitation of planner demonstrations, then by reinforcement learning on a subset of scenes. Randomization strength follows the twin's uncertainty per region and per physical parameter, and twin samples far from a few real samples (held out from the capture and counted in its budget) in a frozen-encoder space [34] are down-weighted. A learned world model (e.g. a video world model [30, 31]) adapted on twin data generates additional trajectories where the twin is most uncertain; it is an ablation without its own threshold.
- *Pre-registered baselines:* uniform randomization of appearance [6] and dynamics [7]; no randomization; ablations without the world model, uncertainty weighting and foundation-model initialization.
- *Criterion (H2):* paired gain ≥ 10 pp on held-out scenes, lower 95% bound > 0 , at ≥ 2 budgets, in the pre-registered regime (baseline $\leq \sim 75\%$ success), 150–300 episodes per arm.

Mechanism 3 – fewer real trials (RQ3, H3; end of sem. 4 – sem. 5).

- *Method:* candidate real trials are scored by the predicted twin-to-reality gap, combining the twin's uncertainty along the planned path, disagreement between twin and world model, and the model's own uncertainty (ensemble spread). The top-ranked trials are run in reality; they correct the twin (re-capture where real and twin observations disagree, counted in the budget; posterior



update of the physical parameters [8, 19]) and the model (co-training with twin data [10]). Real P from few trials is estimated with prediction-powered inference [35].

- *Pre-registered baselines*: random selection; uniform coverage; a failure-driven rule that runs the configurations where the twin predicts failure (as in [18]).
- *Criterion (H3)*: the target P with $\geq 50\%$ fewer trials than random selection (upper 95% bound of the ratio below 1), several runs per arm; against the failure-driven rule only the upper bound below 1 is claimed. Published rules save 20–40% [33]; 50% is expected only where the gap is concentrated.

Whole method (RQ4, H4; sem. 6). Budget curves of the whole method and of the strongest existing approach. No navigation pipeline with a real-trial step exists, so this baseline is *assembled and pre-registered*: capture and fine-tuning as in [12] plus real-trial correction as in [16, 18], with the same reconstruction pipeline, learner, compute and budget; a zero-shot twin and learning from real data only (imitation, same initialization) are reference curves; the method's settings are fixed beforehand. *Criterion (H4)*: $\geq 2\times$ less real data than the assembled baseline at the target P, upper 95% bound of the ratio below 1, and the mechanism 1–3 effects keep their sign; ≥ 30 –50 episodes per point.

Generalization to manipulation (sem. 5–6). The same method, settings unchanged, runs on tabletop tasks in ManiSkill3 [2], where a scene with hidden ground-truth physical parameters and its own rendering is "reality" and the twin is reconstructed from a subset of its views and a few interactions; this weaker proxy is checked against published paired sim-and-real evaluations [5]. No separate thresholds: only budget curves and the sign of the H1–H4 effects.

Real-robot validation. Two campaigns (sem. 5, 7) in 2 PWr rooms with a small mobile robot (TurtleBot 4 Lite from an SzD Minigrant, or K29), ~ 8 robot-hours each; they check direction and the proxy's validity.

Statistics, compute, openness. One-sided tests, $\alpha = 0.05$, scene-level bootstrap, Holm correction; every stage pre-registered. WCSS and PLGrid GPUs. Released: code, configurations, scene lists, seeds and MuSHRoom-based twins; nothing derived from ScanNet++.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / *The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

Streszczenie popularnonaukowe

Roboty, które uczą się zadania na podstawie doświadczenia, na przykład poruszania się po budynku albo przenoszenia przedmiotów, potrzebują wielu takich doświadczeń z miejsca, w którym będą pracować, a zbieranie ich w rzeczywistości jest powolne i kosztowne. Dlatego coraz częściej roboty uczą się w cyfrowym bliźniaku: wiernej kopii prawdziwego miejsca, którą można odtworzyć z krótkiego nagrania i kilku pomiarów. Powstaje w ten sposób droga w trzech krokach: zbieramy dane w rzeczywistości i budujemy z nich kopię, uczymy w niej model robota, a następnie przenosimy go z powrotem do rzeczywistości i tam sprawdzamy oraz poprawiamy. Każdy z tych kroków wymaga jednak prawdziwych danych, a dziś nikt nie wie, jak mało by wystarczyło.

Celem rozprawy jest opracowanie metody, która zmniejsza ilość prawdziwych danych potrzebnych na tej drodze. U jej podstaw leży jedno pytanie naukowe: ile wiedzy o rzeczywistości wnosi każda porcja prawdziwych danych w stosunku do jej kosztu, i kiedy kopia może je zastąpić. Metoda działa na każdym z trzech kroków. W pierwszym, zamiast nagrywać wszystko po równo, nagrywamy przede



wszystkim te miejsca, które są ważne dla zadania, a o których kopia wie jeszcze za mało. W drugim, ponieważ kopia nigdy nie jest idealna, uczymy model tak, aby nie polegał na jej niepewnych fragmentach, a luki w kopii wypełnia program, który uczy się przewidywać, co wydarzy się dalej. W trzecim wybieramy tylko te nieliczne próby w rzeczywistości, które najczęściej mówią o tym, gdzie kopia i model jeszcze się mylą, i wykorzystujemy je do poprawienia obu. Metoda jest opracowywana i sprawdzana na nawigacji robotów mobilnych w budynkach, a następnie, bez zmian, na robotach, które przenoszą przedmioty. Badania są prowadzone przede wszystkim na publicznie dostępnych skanach prawdziwych budynków, a wyniki są potwierdzane na prawdziwym robocie.

Spodziewanym efektem jest metoda, która pozwala nauczyć robota zadania przy dużo mniejszej ilości prawdziwych danych niż dotychczasowe podejścia; to, ile danych oszczędza, mierzymy w minutach pracy człowieka, w ten sam sposób dla każdego podejścia. Może to przyspieszyć i potanić wdrażanie robotów w magazynach, szpitalach i biurach. Opracowane oprogramowanie zostanie udostępnione publicznie.

Abstract for general public

Robots that learn a task from experience, for example moving around a building or handling objects, need a lot of that experience from the place where they will work, and collecting it in reality is slow and costly. That is why robots increasingly learn in a digital twin: a faithful copy of a real place that can be rebuilt from a short recording and a few measurements. This creates a path in three steps: we collect data in reality and build the copy from it, train the robot's model in the copy, and then transfer it back to reality, where we check and correct it. Each of these steps, however, needs real data, and today nobody knows how little would be enough.

The goal of this dissertation is to develop a method that reduces the amount of real data this path needs. At its core lies one scientific question: how much each piece of real data tells us about reality relative to its cost, and when the copy can replace it. The method acts at each of the three steps. In the first, instead of recording everything evenly, we record mainly the places that matter for the task and about which the copy still knows too little. In the second, because the copy is never perfect, we train the model so that it does not rely on the copy's uncertain parts, and a program that learns to predict what happens next fills the gaps in the copy. In the third, we choose only the few trials in reality that tell the most about where the copy and the model are still wrong, and use them to correct both. The method is developed and tested on the navigation of mobile robots in buildings, and then, unchanged, on robots that handle objects. The research is carried out mainly on publicly available scans of real buildings, and the results are confirmed on a real robot.

The expected result is a method that makes it possible to teach a robot a task with much less real data than existing approaches; how much data it saves is measured in minutes of a person's work, in the same way for every approach. This may make deploying robots in warehouses, hospitals and offices faster and cheaper. The software developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of



publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the main track of the **Conference on Neural Information Processing Systems (NeurIPS 2027)**. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports the first reduction mechanism of the method, **less capture** (RQ1, H1), in indoor robot navigation: capture of real data guided by the task and by the twin's uncertainty, evaluated on public scans of real buildings by the real data it saves, in operator minutes, against pre-registered baselines (task T4.2). Because this mechanism is the most exposed to being published first by others, its result is also released as an early preprint. If P1 is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (200-point conferences assigned to the discipline, list of 5.01.2024).

Year	Papers (planned submission, 200 points each)	Code	Grants, licences	Mobility, events
1 (2025/26)	—	—	—	—
2 (2026/27)	P1 (mechanism 1, RQ1): NeurIPS 2027, May 2027 (§11)	Proxy pipeline	ScanNet++; PLGrid; Minigrant; PRELUDIUM; Bekker	Summer school poster
3 (2027/28)	P2 (mechanisms 2–3, RQ2–RQ3, first manipulation results): ICML/CVPR 2028; P3 (whole method, RQ4): NeurIPS 2028	Mechanism 3 code	—	Mid-term; visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Method release	—	Dissertation

Foreign research visit (candidate hosts, not yet contacted): Z. Kira's group, Georgia Tech (EmbodiedSplat [12]); M. Schwager's lab, Stanford (Gaussian-splatting scene models); a European host (Erasmus+) and a foreign co-author for P2 are sought in sem. 4.

Robot and scope. A small mobile robot from an SzD Minigrant or one of the K29 *Denali* laboratory (task T3.2, Nov 2026); static indoor scenes only, moving people are future work.

Risks and mitigation (semester affected).



- *Proxy validity* (3–7): a rendered scan is itself a simulator, and one collision artefact can destroy sim-to-real predictivity [4]; separate captures, a second reference and the sim-vs-real correlation measured against the robot (§9); conclusions are ranking-preserving.
- *Physical gap* (3–6): friction, slip and actuation delays are absent from meshes; "reality" gets hidden physical parameters, mechanism 3 corrects them, the robot checks them.
- *Success ceiling and noise* (3–6): twins already reach high zero-shot success; a pre-registered difficulty regime, hundreds of proxy episodes per arm and targets from the baseline's plateau; robot trials confirm the direction only.
- *No gain from one mechanism, or real-only learning strong at small budgets* (3–6): every curve still answers its RQ, H4 is a full budget curve, and gains from a known location saturate quickly [9].
- *Scooping* (3–6): several groups are one step from a budget curve [12, 18, 32]; H1 is published early (preprint, sem. 4), the literature is re-checked quarterly; we claim the reduction, not the twin.
- *ScanNet++ licence* (3–7): non-commercial, revocable, no redistribution; the supervisor signs; MuSHRoom and own PWR rooms are the releasable fallback; no derived ScanNet++ assets.
- *Compute, robot access* (3–7): standard PLGrid grant and WCSS, smaller models; without a robot, validation moves by a semester (the proxy decides).
- *Rejection* (4–7): resubmission inside the 200-point set 2–4 months later (NeurIPS → CVPR → ICML → ECCV/ICCV).

Prior work, ethics. A pre-PhD NLP paper (ACL 2025 SRW) is background only. Own captures are anonymized (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / **Planned form of cooperation with the supervisor**

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its **% completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Pre-registrations and formal steps.** The supervisor approves the pre-registration of each stage (baselines, thresholds, difficulty regime) before its experiments start, signs the dataset licence application (ScanNet++, Oct 2026) and co-signs the grant applications (§12).
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / **Opinion of assistant supervisor**

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable



.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor